

Secure Settings Apply Options Procedure

Revision

Version 2
9/9/25 2:13 PM

SME

Charles Wilson

Abstract

This document describes the procedure used to perform the security options application activity described in the **AVCDL** ^[1] secondary document **Secure Settings Document** ^[2].

Group / Owner

Security / Security Architect

Motivation

This document is motivated by the need to ensure that the software is built with tools whose selected settings are based on cybersecurity best practices. This is necessary given the nature of safety-critical, cyber-physical systems

Note: Within the context of this document, the terms **security** and **cybersecurity** are used interchangeably. It is presumed that the term **security** is being used in reference to **cybersecurity** and not **physical security**.

License

This work was created by **Torc Robotics** and is licensed under the **Creative Commons Attribution-Share Alike (CC BY-SA-4.0)** License.

<https://creativecommons.org/licenses/by/4.0/legalcode>

Audience

The audience of this document is the cybersecurity practitioner who will be conducting cybersecurity-specific tool setting options activities.

Completeness of Output

Since element-specific tool cybersecurity options application is an as-is analysis, it is critical to ensure that the information gathered is as accurate and complete as possible. As with any cybersecurity activity, it is not the place of the cybersecurity SME to make assumptions on the part of engineering. All information should be sourced from and confirmed by the owner of the element under consideration.

Disposition of Output

Once completed, the generated output should be managed in the organization's document management system (DMS) as a document of record.

Entry Criteria

This document assumes that the reader understands the purpose of cybersecurity-relevant tool settings. Further, that the reader has read and understood the AVCDL **Secure Settings Document** secondary document.

Prerequisites – DevOps SME

Qualifications

It is required that the DevOps SME is both a qualified and trained DevOps SME with expertise in the build system used for the element under consideration in the context of the tool whose cybersecurity settings are being modified.

Note: The specific qualifications and training of DevOps SMEs necessary to assert expertise in providing accurate information regarding the build system used for the element under consideration is outside the scope of this document.

Note: The DevOps group is responsible for determining and making available the appropriate SME for the modifications to build system needed for the tool under consideration.

Knowledge

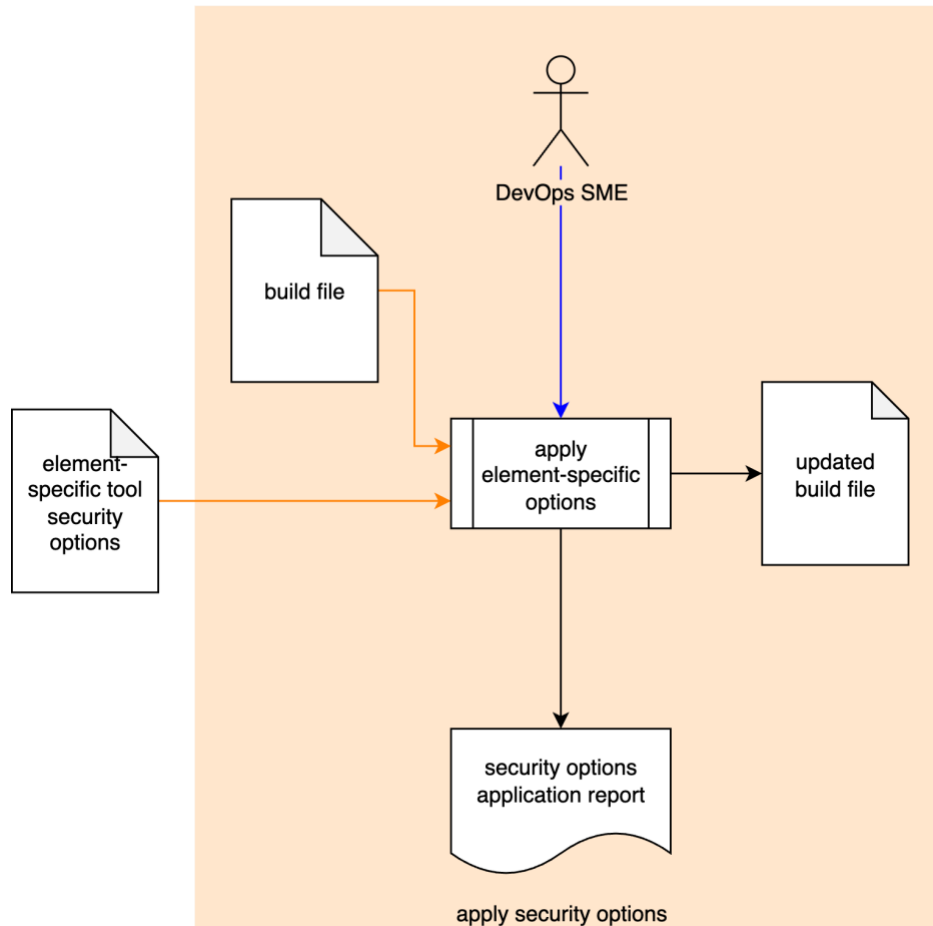
It is required that the DevOps SME have expertise in the build system used for the element under consideration in the context of the tool under whose cybersecurity settings need modification.

Prerequisites – Input Materials

It is required that the DevOps SME acquire all relevant documentation necessary to complete the procedure.

Apply Security Options Activity

The workflow diagram for the **apply security options** activity of the **Secure Settings Document** is shown below.



Using the **Element-specific Tool Security Options** and **Build File**, the DevOps SME applies the selected settings to the appropriate section of the **Build File**. An **Updated Build File** is created and a **Security Options Application Report** is generated.

Methodology

The methodology to be used for selection of element-specific tool cybersecurity options is straightforward. The DevOps SME modifies the build settings as specified in the **Element-specific Tool Security Options**. This is the output from the **Secure Settings – Select Options Procedure** [\[4\]](#) and may be embodied as detailed in the **AVCDL element-specific tool cybersecurity options template** [\[3\]](#).

Note: Depending on the settings added or modified, there may be a negative impact on build time and/or execution speed. These impacts should be accounted for in any timing limiters applied to the build as a whole.

Exit Criteria

This procedure is considered complete once the generated output has been entered into the organization's DMS as a document of record.

Note: The processes and procedures for entering documents into the DMS, or the updating thereof, are outside the scope of this document.

Note: It is recognized that the build file is stored and managed within the context of the revision control system. It is recommended that within regulated domains that an immutable artifact (**Security Options Application Report**) documenting the application of the **Element-specific Tool Security Options** be entered into the DMS for audit purposes.

References

1. **AVCDL** (AVCDL primary document)
2. **Secure Settings Document** (AVCDL secondary document)
3. **AVCDL element-specific tool cybersecurity options template** (AVCDL template)
4. **Secure Settings – Select Options Procedure** (AVCDL tertiary document)